

Prompt Method ^a	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC $\rightarrow$ ReAct	34.2	64.6
ReAct $\rightarrow$ CoT-SC	35.1	62.0
Supervised SoTA ^b	67.5	89.5

Table 1: PaLM-540B prompting results on HotpotQA and Fever.

^aHotpotQA EM is 27.1, 28.9, 33.8 for Standard, CoT, CoT-SC in Wang et al. (2022b).

^b(Zhu et al., 2021; Lewis et al., 2020)

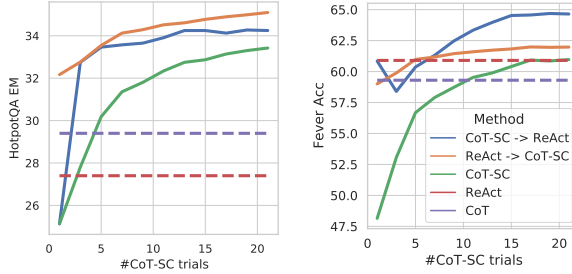


Figure 2: PaLM-540B prompting results with respect to number of CoT-SC samples used.

search reformulation (“maybe I can search/look up x instead”), and synthesize the final answer (“...so the answer is x”). See Appendix C for more details.

Baselines We systematically ablate ReAct trajectories to build prompts for multiple baselines (with formats as Figure 1(1a-1c)): (a) **Standard prompting** (Standard), which removes all thoughts, actions, observations in ReAct trajectories. (b) **Chain-of-thought prompting** (CoT) (Wei et al., 2022), which removes actions and observations and serve as a reasoning-only baseline. We also build a self-consistency baseline (CoT-SC) (Wang et al., 2022a;b) by sampling 21 CoT trajectories with decoding temperature 0.7 during inference and adopting the majority answer, which is found to consistently boost performance over CoT. (c) **Acting-only prompt** (Act), which removes thoughts in ReAct trajectories, loosely resembling how WebGPT (Nakano et al., 2021) interacts with the Internet to answer questions, though it operates on a different task and action space, and uses imitation and reinforcement learning instead of prompting.

Combining Internal and External Knowledge As will be detail in Section 3.3, we observe that the problem solving process demonstrated by ReAct is more factual and grounded, whereas CoT is more accurate in formulating reasoning structure but can easily suffer from hallucinated facts or thoughts. We therefore propose to incorporate ReAct and CoT-SC, and let the model decide when to switch to the other method based on the following heuristics: A) **ReAct $\rightarrow$ CoT-SC**: when ReAct fails to return an answer within given steps, back off to CoT-SC. We set 7 and 5 steps for HotpotQA and FEVER respectively as we find more steps will not improve ReAct performance³. B) **CoT-SC $\rightarrow$ ReAct**: when the majority answer among n CoT-SC samples occurs less than $n/2$ times (i.e. internal knowledge might not support the task confidently), back off to ReAct.

Finetuning Due to the challenge of manually annotating reasoning traces and actions at scale, we consider a bootstrapping approach similar to Zelikman et al. (2022), using 3,000 trajectories with correct answers generated by ReAct (also for other baselines) to finetune smaller language models (PaLM-8/62B) to decode trajectories (all thoughts, actions, observations) conditioned on input questions/claims. More details are in Appendix B.1.

3.3 RESULTS AND OBSERVATIONS

ReAct outperforms Act consistently Table 1 shows HotpotQA and Fever results using PaLM-540B as the base model with different prompting methods. We note that ReAct is better than Act on both tasks, demonstrating the value of reasoning to guide acting, especially for synthesizing the final answer, as shown in Figure 1 (1c-d). Fine-tuning results 3 also confirm the benefit of reasoning traces for more informed acting.

³Of all trajectories with correct final answers, those with 7 steps on HotpotQA and 5 steps on FEVER only take up 0.84% and 1.33% respectively.